Theodoros Mangas

Software Engineer — Python · Django

Komotini, Greece · Remote · open to EU/UK relocation · Employment or contract · theodoros-mangas.github.io/my-portfolio
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PROFILE

Backend-focused Python/Django engineer who builds, ships and runs web applications end to end: data modelling, REST APIs, Celery background jobs, automated testing and Docker-based deployment on Linux. I run two live production platforms, one for a paying client and one for my own engineering practice, and built an engineering SaaS on my own, now in restricted beta with invited users. My background as a licensed land surveyor brings domain expertise in GIS, spatial data and engineering.

EXPERIENCE

Independent Python / Django Developer — Freelance & own products — 2022 – Present · Remote

- Build and maintain Django applications end to end for clients and my own products: data model, backend, interfaces, deployment and ongoing support.
- Run production apps as Docker Compose stacks on Hetzner Cloud: Gunicorn behind a shared Nginx proxy, Let's Encrypt TLS, loopback-only ports, memory limits, and written deploy, backup and restore runbooks.
- **Spec-driven, AI-assisted workflow**: design docs and plans drive coding agents for implementation, tests and security review; I review every change, and CI checks each push.

SELECTED SOFTWARE DELIVERY

HandyTools — Engineering & utilities platform (own product) — Jul 2025 – Present · restricted beta

Django 6 · DRF · Celery · Redis · PostgreSQL 17 · Docker Compose · Nginx · Leaflet

- Built solo (230+ commits): PDF, image and OCR tools plus a full-screen Leaflet **Surveyor Toolkit** for Greek cadastral work, in English and Greek, with Free/Pro/Enterprise tiers enforced server-side via DRF permissions.
- Designed **Celery/Redis** pipelines for OCR and PDF conversion with bounded retries for transient I/O errors, late acknowledgement, time limits and atomic result writes. Uploads up to 50 MB are staged on disk and passed to workers by path, keeping them out of Redis task messages. Fixed a retry bug that discarded successful results and refunded free-tier quota, locked in by a regression test.
- **Spec-first**: larger features start as a design doc and phased plan with explicit scope decisions. **Layered architecture**: framework-free, mypy-checked domain modules behind thin class-based DRF views; coordinate conversions route through one GGRS87 (ΕΓΣΑ87) pivot instead of pairwise transforms.
- **~700 Django tests** in GitHub Actions alongside ruff, mypy and a missing-migration check, with Docker images pushed to GHCR. Greek coordinate transformations (HEPOS, HATT) are tested to the millimetre against official values.
- Per-user quotas enforced under concurrency with **select_for_update** row locks; N+1 queries removed with select_related.

Iasmos Culture Club Museum — Paid client project — Delivered Sep 2026 · live · museum-iasmos-cultureclub.gr

Django 6 · SQLite · Docker · Gunicorn · Nginx · django-modeltranslation

- Bilingual (Greek/English) museum platform with a custom moderator CMS for exhibits, events, pages and media; soft delete with superuser-only restore.
- Privacy by design: self-hosted analytics with daily-rotating visitor hashes, consent-gated YouTube and Maps embeds, moderator Markdown sanitised with nh3, and django-axes login protection.
- **355 automated tests**; documented backup and restore procedures for the database and media.

Agropixida — Engineering practice platform — Live since 2022 · agropixida-syn2.gr

Django · DRF · MySQL · JavaScript · Leaflet · ezdxf · pyproj

- Bilingual business platform with a project showcase and a customer management area.
- Public engineering toolbox: DXF from coordinate files, and DXF/KML/KMZ/GeoJSON conversion with GGRS87 (ΕΓΣΑ87)/WGS84 reprojection.

CityCare — Volunteer team coordination app — Django 6 · PostgreSQL 17 · Leaflet · GitHub Actions

- Teams log actions at mapped locations on a live Leaflet map; per-team daily statistics, rate-limited posts, Greek/English UI. 258 tests in CI, run against both SQLite and PostgreSQL; deployment-ready.

TECHNICAL SKILLS

Backend & data	Python, Django, Django REST Framework, Celery, Redis, PostgreSQL, MySQL, SQLite; ORM optimisation, row locking, rate limiting, django-axes
Design & practice	OOP (class-based views, abstract models, mixins, custom managers, dataclasses); layered architecture; spec-first design docs; AI coding agents (Claude Code)
Delivery & quality	Docker Compose, Linux, Nginx, Gunicorn, Git, GitHub Actions, Django test framework (unittest), ruff, mypy, Sentry
Geo & frontend	JavaScript, HTML/CSS, Bootstrap, Leaflet; pyproj, ezdxf, QGIS; DXF, KML/KMZ, GeoJSON, Shapefile

EARLIER & CONCURRENT EXPERIENCE

Land Surveyor & Geoinformatics Engineer — Independent practice, Greece — 2014 – Present · part-time

Surveying, CAD and GIS work in a multidisciplinary engineering team, plus automation for data processing and validation.

EDUCATION, CERTIFICATION & LANGUAGES

B.Sc. Surveying & Geoinformatics Engineering — TEI of Central Macedonia, 2013
Advanced Python Developer Certificate — PeopleCert, 2022 · Greek (native) · English (fluent)